

Observability-Weighted Terminal Residuals for Nested Krylov Stein Tails

Turning Nonnormal Growth into a Metric Contraction

Liang Wang

July 2026

Abstract

RH-63 repaired the one-step residual failure by recursively expanding Arnoldi defects, but its terminal remainder was still propagated with the Euclidean operator norm. This paper replaces that final factor by a positive Lyapunov metric. For a stable matrix A , let M solve

$$M - A^*MA = I.$$

The induced norm has a strict contraction $\|M^{1/2}AM^{-1/2}\| < 1$. We prove that the coherent nested residual certificate remains valid when its terminal remainder is propagated in this metric, and that it can be multiplied directly into a Stein tail.

On the RH-61 slow/fast surrogate, the weighted certificate is essentially unchanged. On the RH-60 two-block model, the one-level gain drops from 66.41 to 23.84. On the nonnormal four-step chain, whose Euclidean operator norm is 1.110729, the Lyapunov contraction is 0.987369 and the one-level gain drops from 84.05 to 2.485. Full finite-dimensional breakdown remains exact. The new obstruction is metric conditioning: the four-step condition number is about 37.7. No uniform physical-family conditioning theorem, Stage A1 closure, self-adjoint operator, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: Lyapunov metric; weighted residual; nested Krylov; Stein tail; nonnormal operator.

MSC 2020: 47A10; 47B65; 65F35; 93B07; 93D05.

1 Introduction

The directional tail route now has three finite-dimensional layers. RH-61 identified the gap between actual and norm tails. RH-62 projected the source onto an Arnoldi space, and RH-63 recursively projected the residual chain. Both still used an ordinary norm for the terminal residual.

That last choice is structurally poor for nonnormal matrices. A stable matrix can have $\|A\| > 1$ even though its orbit decays. A Lyapunov metric records the decay in a positive norm and is standard in discrete-time stability theory [1]. The question is whether it can be inserted without destroying the coherent nested identity.

Contributions and boundary

1. We construct the positive metric M from the discrete Lyapunov equation and prove strict weighted contraction.
2. We derive the weighted nested terminal-residual certificate.
3. We audit three finite models and quantify the conditioning cost.

All values are deterministic binary64 finite-matrix calculations, with a 256-bit Arb check only for the displayed two-block metric.

2 Weighted residual theorem

Let A be stable and let M be the positive solution of

$$M - A^*MA = I. \quad (1)$$

Define $\|x\|_M = \|M^{1/2}x\|_2$ and

$$q_M = \left\| M^{1/2}AM^{-1/2} \right\|_2. \quad (2)$$

Proposition 2.1 (Strict Lyapunov contraction). *The metric in (1) is positive definite and $q_M < 1$.*

Proof. Stability gives the convergent series $M = \sum_{j \geq 0} (A^*)^j A^j$, hence positivity. From (1),

$$M^{-\frac{1}{2}}A^*MAM^{-\frac{1}{2}} = I - M^{-1}.$$

The largest eigenvalue of the right side is strictly below one. $\square$

Theorem 2.2 (Weighted nested terminal certificate). *Suppose a coherent nested Arnoldi expansion has the form*

$$A^L z = \mathcal{A}_L + \sum_i c_i A^{u_i} g_i.$$

Then

$$\|A^L z\|_M \leq \|\mathcal{A}_L\|_M + \sum_i |c_i| q_M^{u_i} \|g_i\|_M. \quad (3)$$

For a Stein tail $t_L = \sqrt{\kappa} \|A^L z\|_M$, multiplication by $\sqrt{\kappa}$ preserves the upper.

Proof. The Lyapunov similarity gives $\|A^u g\|_M \leq q_M^u \|g\|_M$. Insert this estimate into the coherent decomposition and apply the triangle inequality only after all projected terms have been combined. $\square$

Corollary 2.3 (Conditioning-aware budget). *If $\lambda_{\max}(M) \leq K$ and $\lambda_{\min}(M) \geq k > 0$, then*

$$\|x\|_M \leq \sqrt{K} \|x\|, \quad \|x\| \leq k^{-1/2} \|x\|_M.$$

Thus a weighted residual certificate transfers to Euclidean or Hilbert–Schmidt quantities with a factor controlled by $\sqrt{K/k}$.

Proof. These are the extremal eigenvalue inequalities for a positive matrix. $\square$

Remark 2.4. *The theorem does not assert that the Lyapunov metric is canonical across a noise-dependent physical family. Its finite-dimensional existence is easy; uniform condition numbers and compatibility with Schur packets are the actual analytic questions.*

3 Finite-model audit

The pilot repeats the RH-63 models and compares the Euclidean one-level certificate with the Lyapunov-weighted one at $L = 32$.

Table 1: Weighted terminal residual results.

model	$\ A\ $	q_M	$\text{cond}(M)$	Euclidean gain	weighted gain
slow/fast surrogate	0.993861	0.993861	78.43	1.0000	1.0000
two-block model	0.765889	0.739799	2.13	66.4095	23.8441
four-step chain	1.110729	0.987369	37.66	84.0539	2.4848

The weighted contraction removes the qualitative obstruction $\|A\| > 1$ in the chain. It does not by itself close the tail: the gain remains above one, and the metric condition number is nontrivial.

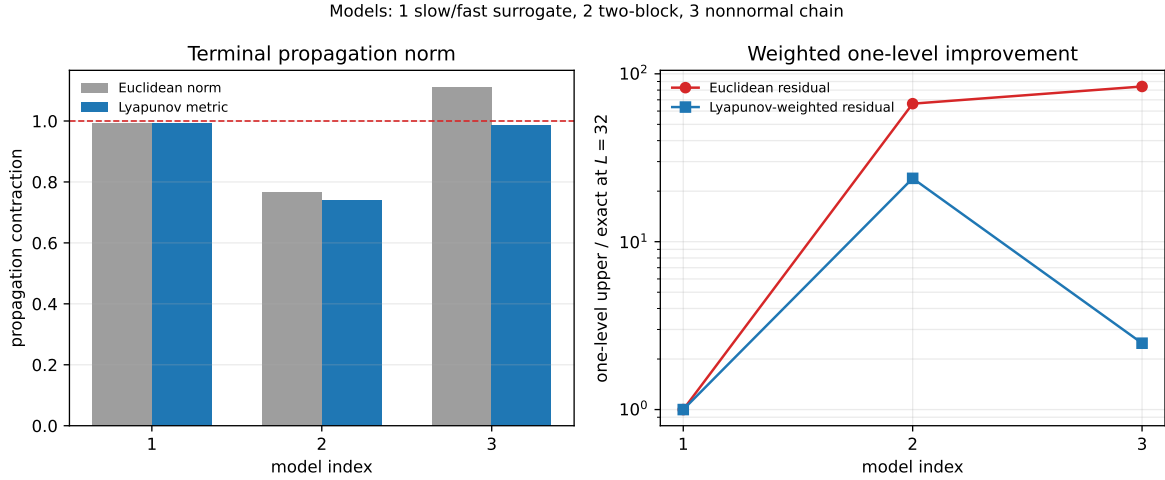


Figure 1: Lyapunov weighting. The left panel compares Euclidean and weighted propagation contractions. The right panel compares one-level residual gains with the unweighted RH-63 certificate.

A 256-bit Arb computation certifies the two-block Lyapunov entries, the identity $M - A^*MA = I$, positivity, and strict weighted contraction. It does not validate the production folded-Gaussian family.

4 Route consequence

The route now has a positive terminal mechanism:

$$\text{nested Krylov projection} \longrightarrow \text{Lyapunov-weighted residual} \longrightarrow \text{Stein tail.} \quad (4)$$

The next gate is not existence of a metric but a family-level ledger:

1. bound the metric condition number on the physical packet prefixes;
2. preserve block cross-column Gram structure;
3. compare the weighted metric with the inherited phase-aware observation metric without double counting conditioning.

If the condition number grows faster than the directional savings, this route becomes a finite-dimensional repair only. If it remains polylogarithmic, RH-64 supplies the missing terminal estimate needed for a physical-family transfer.

5 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no prime-power trace formula, and no completed-zeta identity. It makes no Hilbert–Polya or Riemann-hypothesis claim.

6 Reproducibility

The directory contains the weighted algebra, tests, model pilot, Arb audit, figures, hashes, and publication artifacts. The main commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_weighted_residual_pilot.py
python experiments/run_arb_weighted_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The Lyapunov identities and weighted finite-dimensional certificate are analytic. The model rows are binary64 evidence. Stage A1, Stage A4, uniform metric conditioning, a self-adjoint Hilbert–Polya operator, a prime-power trace formula, and a zeta-zero identity remain open.

References

- [1] Kemin Zhou, John C. Doyle, and Keith Glover. *Robust and Optimal Control*. Prentice Hall, Upper Saddle River, NJ, 1996.